



UK Rail Command Centre

Performance Insights & Analytics using Power BI



Under Supervision: Engineer / Moataz Badran

ANALYTICS TEAM

El Zahraa Saeed Hassan

Merna Samy Aziz

Amira Ezzat Awwad

Sarah Mohamed Abdel Azim

Hayat Nasser Mohamed



TABLE OF CONTENTS

01

Introduction

Project overview, problem definition & objectives

02

Business Requirements

Six key analytical requirement areas

03

Methodology

Research approach, tools & implementation steps

04

Process Overview

Data Groupings & KPIs, Modeling & Star Schema

05

Findings

Key insights, conclusions & recommended actions

INTRODUCTION

Project overview and goals

PROJECT OVERVIEW

This project presents an interactive **Power BI dashboard** developed to analyze UK railway operations performance. It combines Trips (service details), financial & operational data allowing stakeholders to monitor overall network performance through key business indicators such as:

- Total Trips
- Revenue Performance
- On-Time Rate
- Delay and Cancellation Trends
- Average Ticket Value

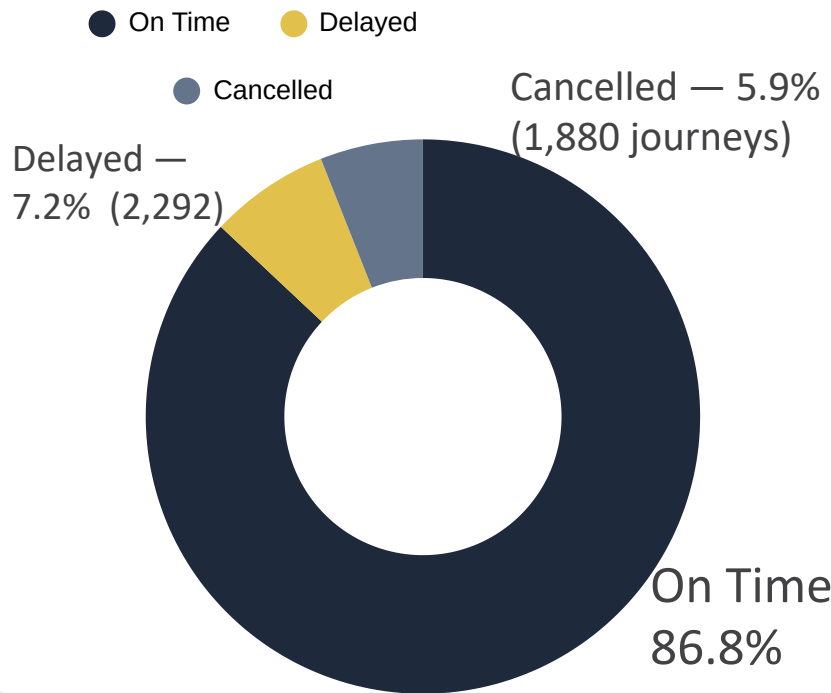
The analysis includes:

- Travel demand distribution across peak hours
- Revenue trends over time
- Ticket type segmentation and customer preferences
- Delay root-cause analysis
- Route-level performance comparison



PROCESS: DATA GROUPINGS & KPIs

JOURNEY STATUS DISTRIBUTION



TICKET TYPE

- Advance — pre-booked, lowest price
- Off-Peak — outside peak hours
- Anytime — flexible, highest price (avg £39.20)

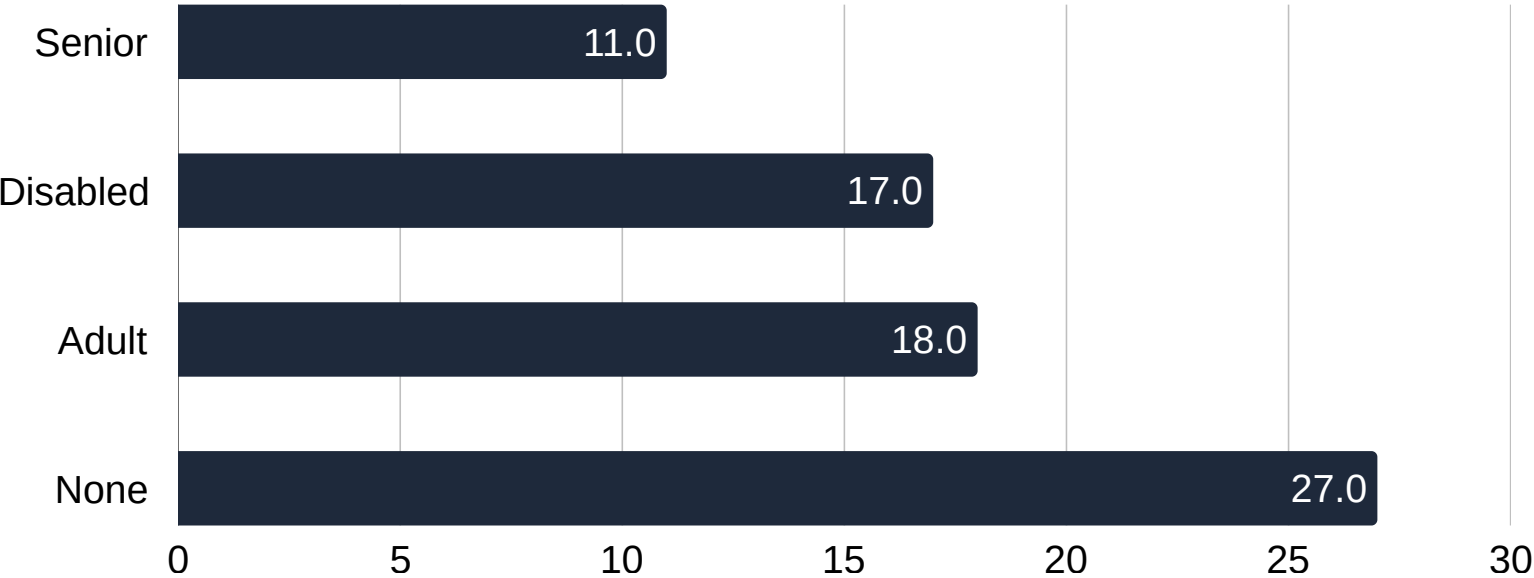
Purchase Type

- Online — 58.5% of rides, avg £20.67
- Station — 41.5% of rides, avg £27.35

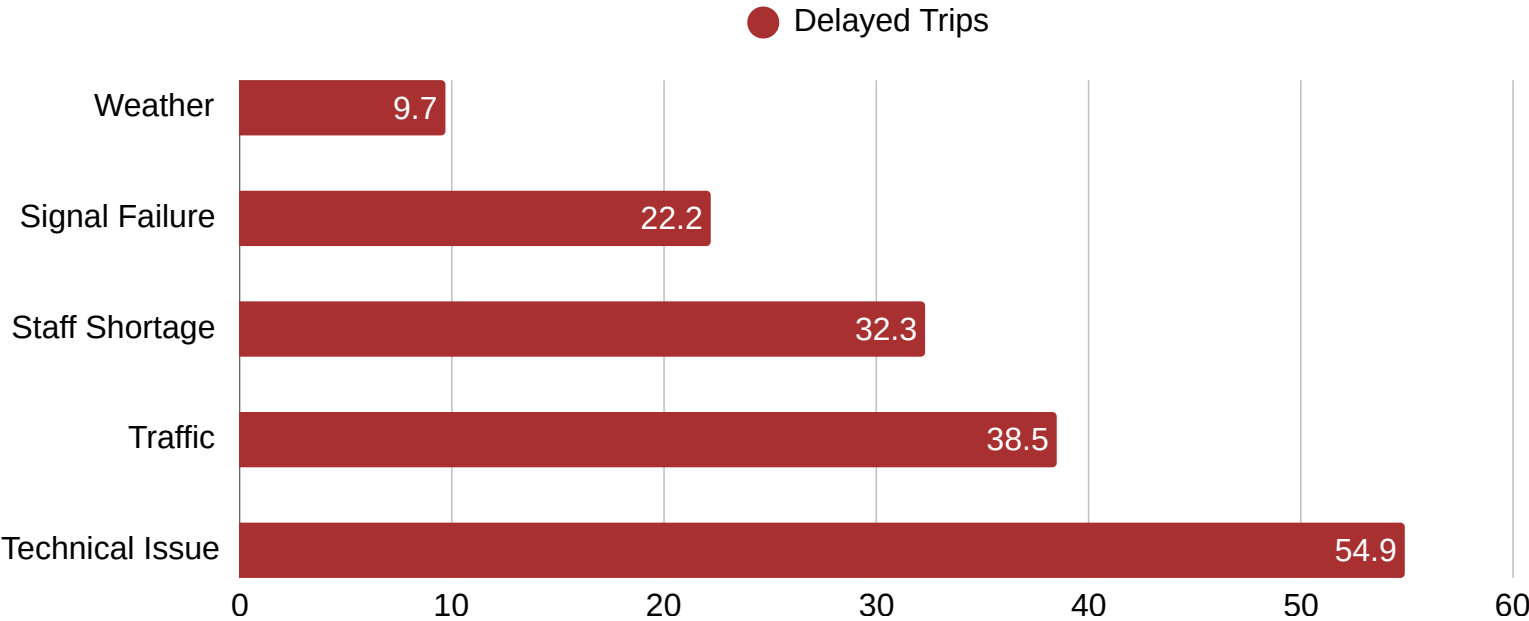
Ticket Class

- Standard — 90.3% of rides
- First Class — 9.7% of rides → 20% of revenue

AVG REVENUE PER RAILCARD TYPE



DELAYED TRIPS BY REASON



INTRODUCTION

PROBLEM DEFINITION

The Challenge

The UK railway operator holds rich transactional data on journeys, yet cannot answer key business questions: Why do revenues dip in certain periods? Which delays drive refund losses? Who are the highest-value customers? What demand is expected next month?



Revenue Leakage

Low-average tickets dominate volume with no upsell mechanism in place



Operational Blind Spot

Delays and cancellations lack root-cause visibility for corrective action



Actions for next Month

No demand forecast means resource planning is reactive rather than proactive



SOLUTION:

- Understand Revenue Sources
- Measure Operational Performance
- Analyse Customer Behaviour
- Forecast Demand & Revenue
- Build Decision-Maker Dashboard

BUSINESS REQUIREMENTS

Service Quality & Satisfaction

- On-time vs delayed trips
- Main delay reasons & routes
- Delay by ticket class
- Refund requests & reasons

Revenue & Sales Performance

- Total revenue & top routes
- Online vs station purchases
- First vs Standard Class revenue
- Revenue by ticket & payment type

Customer Behavior & Segments

- Railcard distribution
- Most active customer segments
- Highest revenue customer groups
- Peak travel times & behaviour

Demand & Route Analysis

- Most / least demanded routes
- Average ticket price per route
- Demand & revenue trends
- Delay trend over time

Time-Based Analysis

- Peak travel hours
- Delays by time of day
- Performance by time period

Insights & Intelligence

- High vs low-performing routes
- Recommendations for improvement



6 Requirement Areas → Covering service quality, revenue performance, customer behavior, demand analysis, time patterns, and business intelligence insights.

Dashboard Views: ● Executive Overview ● Operational ● Financial ● Performance Analysis

METHODOLOGY

Research approach, tools & implementation steps

Research Approach

- Data-driven analytical approach
- Problem Definition
- Data Understanding
- KPI Identification
- Analytical Modeling

Tools & Materials

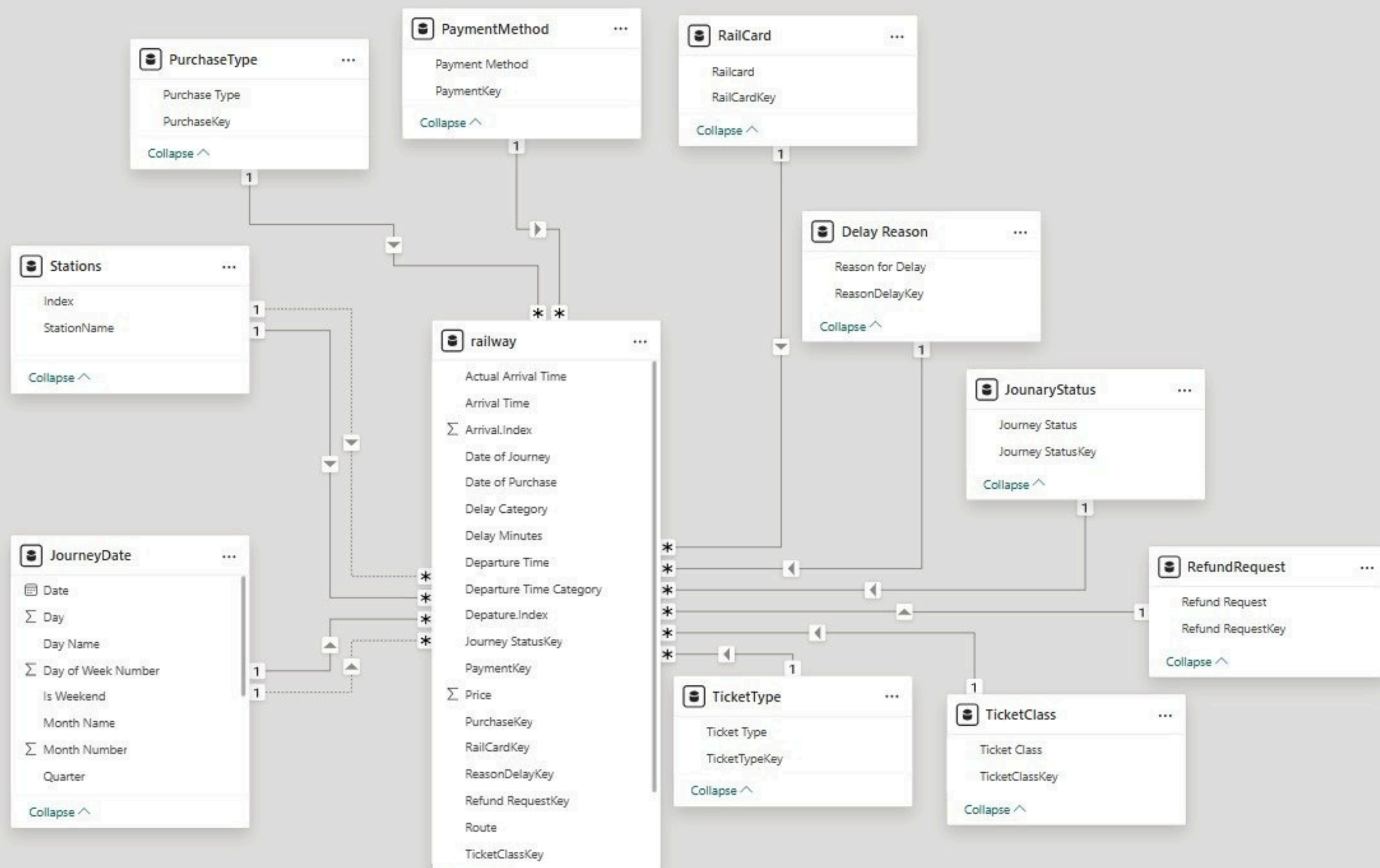
- Power BI Desktop
- Power Query
- DAX (Data Analysis Expressions)
- Star Schema Modeling
- Canva Presentation

Implementation Process

- Step 1: Data Collection
- Step 2: Data Cleaning
- Step 3: Data Modeling
- Step 4: KPI Development
- Step 5: Dashboard Design
- Step 6: Insight Generation

FLOW: Data Collection → Data Cleaning → Data Modeling → KPI Development → Dashboard Design → Insight Generation

PROCESS: DATA MODELING & STAR SCHEMA



FINDINGS — INSIGHTS & ACTIONS

31.65K

Total Trips

£741.92K

Total Revenue

86.82%

On-Time Rate

13.18%

Delay & Cancel %

£23.44

Avg / Trip

1,118

Refunds

FINDING 01

Revenue Loss Distnation

Liverpool Lime Street → London Euston is the highest revenue-loss route with 13.1K in refunds — more than 50% higher than the second route (Manchester Piccadilly → London Euston at 8.7K),

making it the top priority for service reliability improvements.

FINDING 02

Technical delays drive most refunds

Technical issues: **472 trips**, highest refund rate of all categories — more than Weather (**9.7%**).

→ **Prioritise preventive maintenance.** Each fix = fewer refund cases.

FINDING 03

First Class is underutilised

First Class: only **9.7%** of rides yet generates **20%** revenue. Avg **£48.86** vs **£20.72** Standard.

→ **Increase First Class allocation.**

FINDING 04

Morning peak = highest-risk window

Peak travel demand spikes at **hour 6** (3,112 trips) and **hour 18** (3,113 trips), and delays & cancellations follow the exact same pattern — meaning high volume directly drives disruptions.

The recommended action is to redistribute demand away from peak hours through targeted campaigns and off-peak discount incentives

Thank You

UK Rail Command Centre · Power BI Analytics Dashboard

ANALYTICS TEAM

El Zahraa Saeed Hassan · Merna Samy Aziz · Sarah Mohamed Abdel Azim
Amira Ezzat Awwad · Hayat Nasser Mohamed

Under Supervision of **Engineer / Moataz Badran**

